This document provides additional assistance with wiring your Extron IP Link Pro Control Processor to your device. Different components may require a different wiring scheme than those listed below.

For complete operating instructions, refer to the user's manual for the specific IP Link Pro Control Processor or the documentation supplied by the manufacturer of the controlled device.

For more information on using Global Scripter Modules, refer to the "Guide to Using Scripter Modules" document.

Device Specifications

Device Type:	Video Projector
Manufacturer:	Barco
Firmware Version:	N/A
Model(s):	RLM-W8

Tested on the Following Software and Firmware Versions

IP Link Pro Control Processor Firmware	Global Scripter Version
2.06.0000-b010	1.4.1

Version History

Module Version	Date	Notes
1_0_1_0	11/8/2017	Initial Version

Module Notes

- Unidirectional variable must be set to 'True' if status is not required. Default value is 'False'.
  Example: `InterfaceName.Unidirectional = 'True'`

- connectionCounter variable must be set to the number of queries that will be sent to the device before displaying 'Disconnected' if no response is received. Default value is 15.
  Example: `InterfaceName.connectionCounter = 5`

Supported Classes and Examples

SerialClass

```
InterfaceName = ModuleName.SerialClass(ProcessorName, 'COM1', Model='RLM-W8')
```

SerialOverEthernetClass

```
InterfaceName = ModuleName.SerialOverEthernetClass('192.168.254.254', 2001, Model='RLM-W8')
```

EthernetClass

```
InterfaceName = ModuleName.EthernetClass('192.168.254.254', 43680, Model='RLM-W8')
```

Set Commands

Format with Qualifier:

```
InterfaceName.Set(Command, Value, {'Qualifier Key': 'Qualifier Value'})
```

Format without Qualifier:

```
InterfaceName.Set(Command, Value)
```

Command	Value	Value	Value
AspectRatio	'5:4'	'4:3'	'16:10'
	'16:9'	'1.88'	'2.35'
	'Letterbox'	'Native'	'Unscaled'

```
# AspectRatio example
InterfaceName.Set('AspectRatio', '5:4')
```

Command	Value
AutoImage	None

```
# AutoImage example
InterfaceName.Set('AutoImage', None)
```

Command	Value	Value	Value
Input	'HDMI 1'	'HDMI 2'	'RGB'
	'YUV 1'	'YUV 2'	'Composite'
	'S-Video'	'RGBS'	'HD SDI'

```
# Input example
InterfaceName.Set('Input', 'HDMI 1')
```

Command	Value	Value	Value
LampMode	'Eco'	'Standard'	'Low'

```
# LampMode example
InterfaceName.Set('LampMode', 'Eco')
```

Command	Value	Value
LampSelect	'Single'	'Dual'

```
# LampSelect example
InterfaceName.Set('LampSelect', 'Single')
```

Command	Value	Value
Power	'On'	'Off'

```
# Power example
InterfaceName.Set('Power', 'On')
```

Command	Value	Value
VideoMute	'On'	'Off'

```
# VideoMute example
InterfaceName.Set('VideoMute', 'On')
```

Status Available

For all commands, call Update to receive the latest status. ConnectionStatus, DeviceStatus do not support the Update function and is triggered by the device providing a successful response to other Update function calls.

Format with Qualifier:

```
InterfaceName.Update(Command, {'Qualifier Key': 'Qualifier Value'})
Value = InterfaceName.ReadStatus(Command, {'Qualifier Key': 'Qualifier Value'})
InterfaceName.SubscribeStatus(Command, {'Qualifier Key': 'Qualifier Value'}, FeedbackHandler)
```

FeedbackHandler will be called only when the specified qualifier gets a new status.

Format without Qualifier:

```
InterfaceName.Update(Command)
Value = InterfaceName.ReadStatus(Command)
InterfaceName.SubscribeStatus(Command, None, FeedbackHandler)
```

FeedbackHandler will be called when any qualifier gets a new status.

Command	Value	Value	Value
AspectRatio	'5:4'	'4:3'	'16:10'
	'16:9'	'1.88'	'2.35'
	'Letterbox'	'Native'	'Unscaled'

```
# AspectRatio examples
InterfaceName.Update('AspectRatio')
Value = InterfaceName.ReadStatus('AspectRatio')
InterfaceName.SubscribeStatus('AspectRatio', None, FeedbackHandler)
```

Command	Value	Value
ConnectionStatus	'Connected'	'Disconnected'

```
# ConnectionStatus examples
Value = InterfaceName.ReadStatus('ConnectionStatus')
InterfaceName.SubscribeStatus('ConnectionStatus', None, FeedbackHandler)
```

Command	Value	Value	Value
DeviceStatus	'Inlet Temp Over'	'DMD Error'	'Lamp Overheat'
	'Lamp Ballast Overheated'	'Fan Error'	'Lamp does not ignite'
	'Lamp Ignition Failure'	'Ballast Communication Error'	'GPIO Failure'
	'Interlock'	'No Signal Present'	'System Error'
	'Software I2C Failure'	'EEPROM Failure'	'EDID Failure'
	'EEP Version Failure'	'RST Gennum'	'Temp Sensor Failure'
	'System Failure'	'Normal'	

```
# DeviceStatus examples
Value = InterfaceName.ReadStatus('DeviceStatus')
InterfaceName.SubscribeStatus('DeviceStatus', None, FeedbackHandler)
```

Command	Value	Value	Value
Input	'HDMI 1'	'HDMI 2'	'RGB'

	'YUV 1'	'YUV 2'	'Composite'
	'S-Video'	'RGBS'	'HD SDI'

```
# Input examples
InterfaceName.Update('Input')
Value = InterfaceName.ReadStatus('Input')
InterfaceName.SubscribeStatus('Input', None, FeedbackHandler)
```

Command	Value	Value	Value
LampMode	'Eco'	'Standard'	'Low'

```
# LampMode examples
InterfaceName.Update('LampMode')
Value = InterfaceName.ReadStatus('LampMode')
InterfaceName.SubscribeStatus('LampMode', None, FeedbackHandler)
```

Command	Value	Value
LampSelect	'Single'	'Dual'

```
# LampSelect examples
InterfaceName.Update('LampSelect')
Value = InterfaceName.ReadStatus('LampSelect')
InterfaceName.SubscribeStatus('LampSelect', None, FeedbackHandler)
```

Command	Value	Value
LampStatus	'On'	'Off'
Qualifier Key	Qualifier Value	
'Lamp'	'1' – '2'	

```
# LampStatus examples
InterfaceName.Update('LampStatus', {'Lamp': '1'})
Value = InterfaceName.ReadStatus('LampStatus', {'Lamp': '1'})
InterfaceName.SubscribeStatus('LampStatus', None, FeedbackHandler)
```

Command	Value
LampUsage	0 –
Qualifier Key	Qualifier Value
'Lamp'	'1' – '2'

```
# LampUsage examples
InterfaceName.Update('LampUsage', {'Lamp': '1'})
Value = InterfaceName.ReadStatus('LampUsage', {'Lamp': '1'})
InterfaceName.SubscribeStatus('LampUsage', None, FeedbackHandler)
```

Command	Value
OperationHours	0 –

```
# OperationHours examples
InterfaceName.Update('OperationHours')
Value = InterfaceName.ReadStatus('OperationHours')
InterfaceName.SubscribeStatus('OperationHours', None, FeedbackHandler)
```

Command	Value	Value	Value
Power	'On'	'Off'	'Warming'
	'Cooling'	'Warning'	

```
# Power examples
```

```
InterfaceName.Update('Power')
Value = InterfaceName.ReadStatus('Power')
InterfaceName.SubscribeStatus('Power', None, FeedbackHandler)
```

Command	Value	Value
VideoMute	'On'	'Off'

```
# VideoMute examples
InterfaceName.Update('VideoMute')
Value = InterfaceName.ReadStatus('VideoMute')
InterfaceName.SubscribeStatus('VideoMute', None, FeedbackHandler)
```

Cable and Adapter Requirements

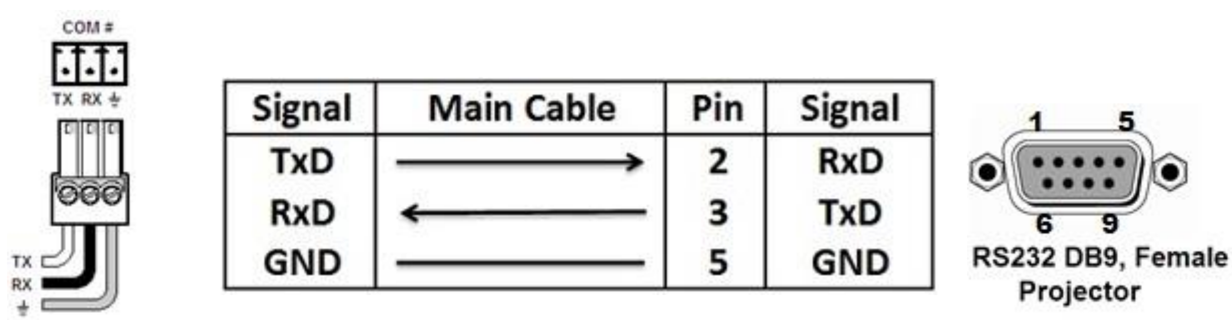
Captive Screw to Male DB9 RS-232 Serial Cable

Notes for the Device

Serial communication

Port Type:	RS-232	**Parity:**	None
Baud Rate:	38400	**Stop Bits:**	One
Data Bits:	8	**Flow Control:**	None

Pin Assignments Diagram



Network communication

When configuring the Ethernet module, be sure device settings match those of the Global Scripter ethernet interface

Port Type:	Ethernet
Default Port:	43680
Logon Credentials Supported:	No
Multi-Connection Capabilities:	Undetermined
Port Changeability:	Undetermined

Ethernet Module Configuration Description

Please refer to user manual for settings and changes to the network communication

Notes for the Device

Appendix A. Set Commands

Aspect Ratio 1.88	op aspect = 4\x0D
Aspect Ratio 16:10	op aspect = 2\x0D
Aspect Ratio 16:9	op aspect = 3\x0D
Aspect Ratio 2.35	op aspect = 5\x0D
Aspect Ratio 4:3	op aspect = 1\x0D
Aspect Ratio 5:4	op aspect = 0\x0D
Aspect Ratio Letterbox	op aspect = 6\x0D
Aspect Ratio Native	op aspect = 7\x0D
Aspect Ratio Unscaled	op aspect = 8\x0D
Auto Image None	op auto.img\x0D
Input Composite	op input.sel = 5\x0D
Input HD SDI	op input.sel = 8\x0D
Input HDMI 1	op input.sel = 0\x0D
Input HDMI 2	op input.sel = 1\x0D
Input RGB	op input.sel = 2\x0D
Input RGBS	op input.sel = 7\x0D
Input S-Video	op input.sel = 6\x0D
Input YUV 1	op input.sel = 3\x0D
Input YUV 2	op input.sel = 4\x0D
Lamp Mode Eco	op lamp.mode = 0\x0D
Lamp Mode Low	op lamp.mode = 2\x0D
Lamp Mode Standard	op lamp.mode = 1\x0D
Lamp Select Dual	op lamps = 1\x0D
Lamp Select Single	op lamps = 0\x0D
Power Off	op power.off\x0D
Power On	op power.on\x0D
Video Mute Off	op picture.mute = 0\x0D
Video Mute On	op picture.mute = 1\x0D

Appendix B. Query Commands

Aspect Ratio	op aspect ?\x0D
Device Status	op errcode ?\x0D
Input	op input.sel ?\x0D
Lamp Mode	op lamp.mode ?\x0D
Lamp Select	op lamps ?\x0D
Lamp Status Lamp 1	op lamp1.stat ?\x0D
Lamp Status Lamp 2	op lamp2.stat ?\x0D
Lamp Usage Lamp 1	op lamp1.hours ?\x0D
Lamp Usage Lamp 2	op lamp2.hours ?\x0D
Operation Hours	op proj.runtime ?\x0D
Power	op status ?\x0D

Video Mute	op picture.mute ?\x0D